Sayma Islam

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Professional Summary

Robotics and Mechatronics Engineering researcher specializing in soft robotics, compliant mechanisms, structural finite element analysis (FEA), and bio-inspired mechanical synthesis. Experienced in full-cycle hardware design and physical prototyping—spanning smart origami structures, flexible dynamic materials, multi-material additive manufacturing, non-linear deformation analysis, and embedded control integration. Proven track record in parametric CAD modeling, ROS2 dynamics, and hardware validation for adaptive robotic mechanisms.

Education

- **B.Sc. in Robotics & Mechatronics Engineering** 2022 – Expected November 2026
  University of Dhaka *Dhaka, Bangladesh*

  Undergraduate Thesis (2025–26): Developing an origami-inspired inchworm robot for in-pipe maintenance featuring minimal actuation, compliant origami structures, and proprioceptive terrain awareness.
  Key Research Focus: Synthesis of underactuated dynamic structures, kinematic deformation under geometric constraints, compliant mechanism design, flexible material stress distribution, and soft sensor integration.
  Supervisors: Dr. Shamim Ahmed Deowan, Dr. Shugata Ahmed **Contact:** rme@du.ac.bd

Publications

1. S. J. Saha, M. A. Chowdhury, **S. Islam**, S. A. Deowan, S. E. Arman, and A. K. Ghosh, "Flexipede: A Bio-Inspired, Modular Myriapod Robot for Rough-Terrain Traversal," *Robotics* (MDPI), 2026. [**Q1 Journal** — DOI: 10.3390/robotics15070129]
2. S. J. Saha, M. Akter, N. Jahan, **S. Islam**, A. Chowdhury, M. A. Shabur, "Material Selection and Fiber Orientation Effects in Composite Overwrapped Steel Pipelines: A Numerical Study for Burst Pressure Enhancement," *Journal of Sustainable Smart Materials and Structural Systems*, vol. 2, pp. 61–74, 2026. [Link]
3. M. Akter, A. B. Spriha, **S. Islam**, N. Jahan, N. Z. Rahman, S. E. Arman, and L. Jamal, "Not a Replacement, but a Companion: The Paradigm Shift in AI-Assisted Learning Among Engineering Students in Bangladesh," in *Proc. 14th Int. Conf. on Information and Education Technology (ICIET 2026)*, Koriyama, Japan, 2026. [Link]

Manuscripts

1. A. I. Hemel, M. Akter, **S. Islam**, M. M. Mugdha, A. B. Spriha, and S. A. Deowan, "AURIS: An Edge-Oriented Framework for Structural Health Monitoring and Physical Fingerprinting Using High-Frequency Vibration Sensing," *Submitted for Journal Publication (Under Review)*, 2026.

Work Experience

- **BioDynamics Lab, University of Dhaka** Dhaka, Bangladesh
  Undergraduate Researcher – Supervisor: Dr. Shamim Ahmed Deowan *2025 – Present*
  - Undergraduate thesis and further research on origami-inspired and compliant robotic mechanisms.
- **University of Dhaka** Dhaka, Bangladesh
  Research Assistant – Geckobot Project *Sep 2024 – Nov 2025*
  - **Supervisors:** Dr. Shamim Ahmed Deowan, Dr. Mosaddek Hossain Kamal Tushar
  - Engineered bio-inspired compliant adhesion mechanisms and soft contact interfaces for **Geckobot** (funded by ICT Division, Govt. of Bangladesh).
  - Developed high-fidelity CAD models and dynamic ANSYS stress-deformation simulations to optimize dynamic adhesion force distribution and flexibility.
  - Fabricated physical prototypes using multi-material 3D printing (FDM) and led experimental validation setup for soft contact mechanics and climbing performance.
- **Mechatronics and Artificial Intelligence in Medicine (MAIM) Lab** Dhaka, Bangladesh
  Undergraduate Research Assistant – Director: Dr. Abhishek Kumar Ghosh *Nov 2024 – Mar 2025*
  - Spearheaded mechanical design and parametric 3D CAD modeling for the **Flexipede** bio-inspired robot; synthesized modular compliant joints and structural interconnects optimized for rough-terrain traversal.
  - Conducted non-linear finite element analysis (FEA) in ANSYS to evaluate structural stress distribution, dynamic deformation, and fatigue limits under dynamic terrain loads.
  - Executed full-cycle hardware development, managing multi-material FDM additive manufacturing, precision metrology, BOMs, and component sourcing for physical dynamic prototypes.

- **Cortex AI Lab** Dhaka, Bangladesh
  Researcher – Director: Shifat E. Arman *Feb 2024 – Feb 2025*
  - Modeled multi-body forward/inverse kinematics and non-linear structural compliance for soft-rigid hybrid bio-inspired locomotion mechanisms.
  - Analyzed dynamic gait stability and flexible interconnect deformation mechanics to optimize underactuated locomotion control on unstructured surfaces.
- **Technyx** Dhaka, Bangladesh
  Product Designer *April 2025*
  - Developed high-fidelity parametric 3D CAD models, assembly renders, and structural geometry using Fusion 360 for industrial applications.

Research Projects

- **Reinforcement Learning Frameworks for Soft & Flexible Robotics**
  Ongoing Research Project
  - Developing RL control environments and compliant mechanism policies for underactuated soft robotic manipulators.
- **Burst Pressure Enhancement of Steel Pipelines Using Variable-Angle Composite Overwraps**
  Numerical FEA Study – Supervisor: Dr. Md. Abdus Shabur *Oct 2025 – Feb 2026*
  - Modeled anisotropic material mechanics, interlaminar stress, and non-linear deformation dynamics in ANSYS ACP for composite-wrapped structures.
- **Airbus Fly Your Ideas 2026 Challenge**
  Global Technical Competition – Team Skyrim *2026*
  - Engineered parametric CAD concepts and executed structural stress validation in ANSYS for next-generation aerospace structures.
- **Autonomous Navigation & Kinematic URDF Modeling**
  ROS2 Project *2026*
  - Modeled dynamic URDF structures and autonomous navigation environments using ROS2 Jazzy.
- **Ergonomic Compliant Thumb-Wrist Splint Design**
  Technical Research Project *2024*
  - Designed, simulated, and 3D printed a flexible, bio-compliant wrist splint for custom orthopedic stabilization.

Technical Skills

- **Soft Robotics & Bio-Inspired Mechanics:** Compliant Mechanism Design, Biomimetic Kinematics, Underactuated Systems, Soft-Rigid Hybrid Architectures, Flexible Structural Interconnects
- **Computational Design & CAE:** Parametric 3D CAD Modeling (Fusion 360, SolidWorks), Finite Element Analysis (ANSYS ACP, Static Structural), COMSOL Multiphysics, Non-linear Material Deformation, GD&T
- **Mathematical Modeling & Programming:** Python, C++, MATLAB, Kinematic & Dynamic Modeling, Geometric Computing Concepts, Reinforcement Learning (RL) Frameworks
- **Robotics Middleware & Software Tools:** ROS2 (Jazzy), URDF Structural & Dynamic Modeling, KiCad PCB Design, Git/GitHub Version Control
- **Prototyping, Manufacturing & Metrology:** Multi-Material Additive Manufacturing (FDM 3D Printing), CAM/G-code, Precision Metrology (Calipers, Multimeters), Experimental Kinematic Validation & Physical Transfer

Achievements

- **Silver Medal — World Invention Competition & Exhibition (WICE)**
  Flexipede project; selected for international finals in Malaysia
- **4th Place — Global Robotics & Innovation Consortium (GRIC)**
  Flexipede project; selected for international finals in Turkey
- **Round 2 Qualifier — Airbus Fly Your Ideas Global Challenge** 2026
  Team Skyrim; selected among global entrants for advanced proposal evaluation

Interests

Deeply passionate about soft robotics, compliant mechanism design, smart origami metamaterials, and bio-inspired actuation. Drawn to the synthesis of flexible dynamic structures and novel hardware fabrication techniques. Also interested in observational astrophysics.